Water Circulation & Productivity in Possession Sound

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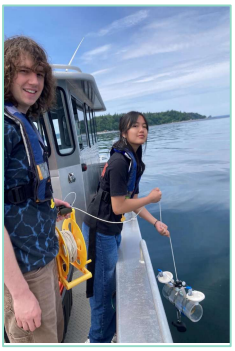
Ocean Research College Academy
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Introduction

The food web within Possession Sound is reliant on phytoplankton (primary producers), which can be estimated by measuring chlorophyll. Circulation in Possession Sound is led by inputs of fresh river water from the Snohomish River and seawater from the surrounding straits. Primary productivity is affected by the circulation of water in the estuary as primary producers are moved toward or away from the sunlit surface waters and as the water conditions change. Certain areas of Puget Sound, including the Possession Sound region, have been found to have lower water circulation compared to other areas.[1] In addition, surface water salinity in Possession Sound is more affected by freshwater inputs while deeper waters largely influenced by the circulating seawater.[2] Chlorophyll has also been found to be highly variable and complex, needing more research to be fully understood.[3] This study aims to better understand the influence of circulation on productivity. Based on previous studies, we hypothesized that higher river discharge will increase chlorophyll as freshwater carries nutrients into Possession Sound. Conversely, high tides and increased salinity could lower the amount of chlorophyll as salinity increases stratification, decreasing the amount of mixing and potentially keeping the nutrients from reaching the phytoplankton.

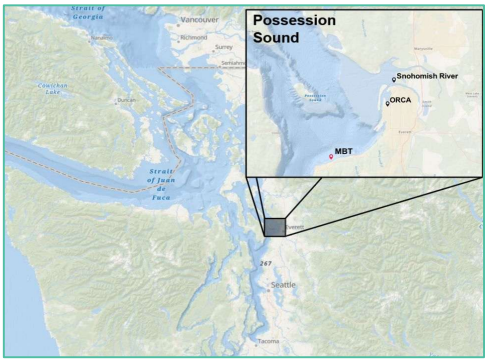






Research Question: How is productivity influenced by circulation in Possession Sound?

Study Site



Methods

To collect data, a YSI Exo2 Sonde was deployed at Mount Baker Terminal (MBT), where chlorophyll and salinity profiles were collected through the water column. To obtain tide and river discharge data, NOAA and USGS datasets were utilized.[4-5]





Results

Chlorophyll levels during **high tide exhibited a larger range** (0-18.27 RFU) than it did during low tide (0.02-11.85 RFU). However, max chlorophyll during high tides compared to low tides showed no statistically significant difference between groups (p=0.84). We also observed a **greater standard deviation of mean chlorophyll during higher salinities,** and river discharge showed **greater standard deviation of mean chlorophyll during dates with higher river discharge.** Similarly, there was no significant difference between these data points as all error bars overlapped.

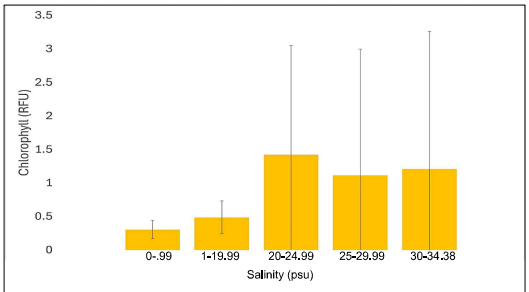


Figure 1: Mean chlorophyll grouped by different ranges of salinity. Error bars represent standard deviation of the mean.

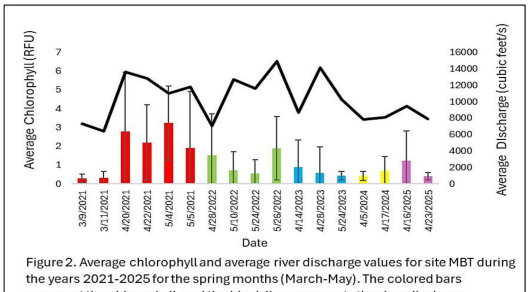


Figure 2. Average chlorophyll and average river discharge values for site MBT during the years 2021-2025 for the spring months (March-May). The colored bars represent the chlorophyll, and the black line represents the river discharge.

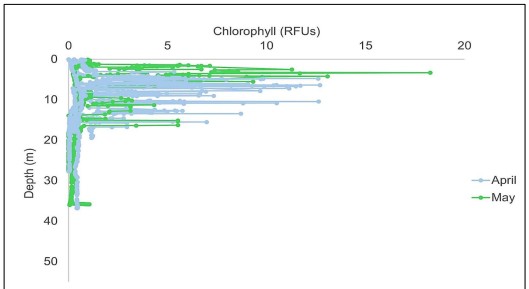


Figure 3. YSI EXO2 Sonde chlorophyll water column profiles collected at MBT (Mount Baker Terminal) between 2020-2025. Above is every profile collected during spring high tides.

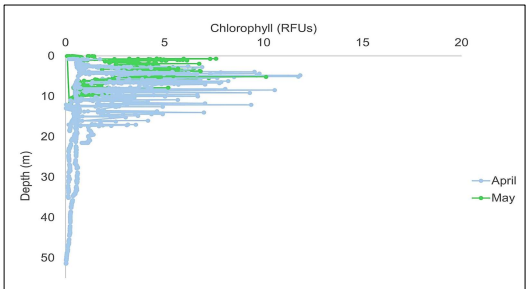


Figure 4.YSI EXO2 Sonde chlorophyll water column profiles collected at MBT (Mount Baker Terminal) between 2020-2025. Above is every profile collected during spring low tides.

Conclusions

- The impact of water circulation on primary productivity is highly complex. Possession Sound is a dynamic estuary with influxes of nutrients from runoff and anthropogenic sources; thus, nutrients may not always be a limiting factor to productivity.
- The increase in chlorophyll associated with higher salinities was possibly due to increased stratification, but this difference was not significant. It is important to note that this estuary has a halocline year-round that prevents vertical mixing and influences where plankton settle in the water column.
- Increased range of chlorophyll during higher tides and river discharge could be caused by freshwater affecting stratification, turbidity blocking sunlight, or the circulation of nutrient rich waters towards or away from the sample site (MBT).
- The higher stratification in Possession Sound, in comparison to the wider Salish Sea, may play an important role in why there are no significant changes in productivity with changing river discharge or tides, as we did not observe these factors to have a significant effect in our study.[1]
- This estuary is highly complex. Thus, further research and dynamic modeling should be carried out in order to better and understand the effects of changing water circulation on chlorophyll and productivity. It is also imperative to consider movements of nutrients during high tides and river discharge in comparison to chlorophyll levels, as well as to measure amounts of sunlight and turbidity levels.

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